

RE VALUEWORKS

Real Estate Equity & Asset-Management Analytics Platform
Investment Committee Memorandum

Prepared by Jude Christina Gaspard

July 2026

Property-level data is synthetic. Public REIT reference figures are real and cited. See Data Dictionary for full disclosure.

1. Executive Summary

RE ValueWorks is an equity and asset-management analytics platform built to give a real estate investment team continuous, scored visibility into a 250-property portfolio: which assets to hold, which need repositioning capital, and which have already created enough value to sell. This memo summarizes the portfolio's current state, the forecasting methodology behind it, and how returns move under a defined downside scenario.

Portfolio snapshot at baseline:

- Implied portfolio value: \$9,469M
- Total equity invested: \$3,481M
- Median IRR: 8.3% | Median equity multiple: 1.33x
- Recommendation mix: 122 Hold, 94 Sell, 34 Reposition (of 250 properties)
- 66 properties (26.4%) are currently impaired — negative or undefined IRR

Office is the single largest risk concentration: the only property type with a negative average IRR (−6.3%), driven by cap rate expansion since acquisition — consistent with the real, observable pattern in public office REITs since 2022 (see Section 4). A defined moderate-downside scenario (−5% rent, −8% occupancy, +75bps exit cap rate, +15% capex) pushes median portfolio IRR from +8.3% to −8.8% and impaired properties from 66 to 186 — more than doubling the share of the book in distress.

2. Business Problem & Approach

A real estate investment manager overseeing a multi-property portfolio needs a continuous answer to six questions: which assets are outperforming, which require capital intervention, which markets show improving fundamentals, where occupancy is weakening, whether an asset should be held, renovated, repositioned, or sold, and how financing costs affect equity returns.

RE ValueWorks answers these with four layers:

- A documented synthetic property portfolio, standing in for proprietary fund-level data
- A real cash-flow-based IRR/equity multiple model per property, plus unsupervised segmentation and a forward NOI growth forecast
- An interactive Streamlit application for live scenario/sensitivity analysis
- A Power BI dashboard for investment-committee reporting

The model does not decide whether to buy, sell, or hold an asset. It structures the investment discussion — the final call remains a human decision, consistent with the project's design principle.

3. Data & Methodology

3.1 Data

The 250-property portfolio is synthetic — fund-level PE real estate data is proprietary and not published. Unlike a purely random dataset, every property's IRR and equity multiple are the actual output of a levered cash-flow model (equity invested, annual cash flow, terminal sale proceeds), not independently assigned figures. This is enriched with a separate, real reference layer: manually compiled, cited financial data for 10 large-cap public REITs (reports/reit_reference_benchmarks.md), used as a directional sanity check rather than a statistical calibration.

3.2 Modeling

Two genuinely separate exercises: unsupervised KMeans segmentation (four performance-based clusters), and a supervised Random Forest forecast of next year's same-property NOI growth. IRR and equity multiple were deliberately NOT modeled with machine learning — in this dataset they are exact formulas with no noise term, so “predicting” them from the same inputs that generated them would just be reverse-engineering algebra, not real forecasting.

3.3 Forward NOI growth forecast — held-out test results

Metric	Value
R²	0.611
MAE	1.10 percentage points
Naïve baseline MAE (persistence forecast)	1.52 percentage points

The model beats the naive “next year equals this year” baseline (1.10 vs. 1.52 points of error) — the bar that actually matters, since a forecast that doesn't beat doing nothing isn't earning its complexity. An R^2 of 0.61 is reported as-is rather than implying more certainty than a one-year NOI forecast can honestly claim.

4. Portfolio Segmentation & Findings

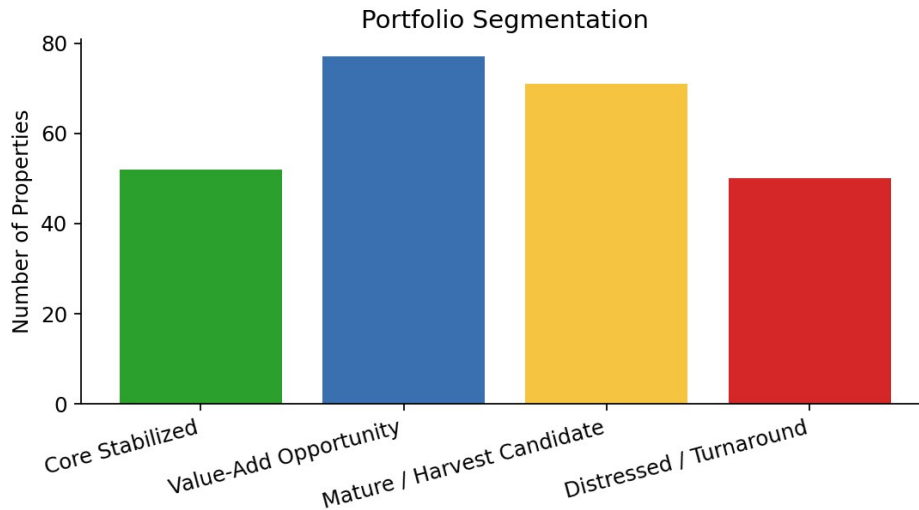


Figure 1. Four segments emerge from unsupervised clustering on occupancy, NOI growth, leverage, interest coverage, and NOI margin — not manually defined.

4.1 Segment characteristics

Segment	Avg NOI Growth / Avg Leverage
Core Stabilized (52 properties)	+3.9% / 55%
Value-Add Opportunity (77 properties)	+3.3% / 66%
Mature / Harvest Candidate (71 properties)	+3.0% / 61%
Distressed / Turnaround (50 properties)	-1.5% / 63%

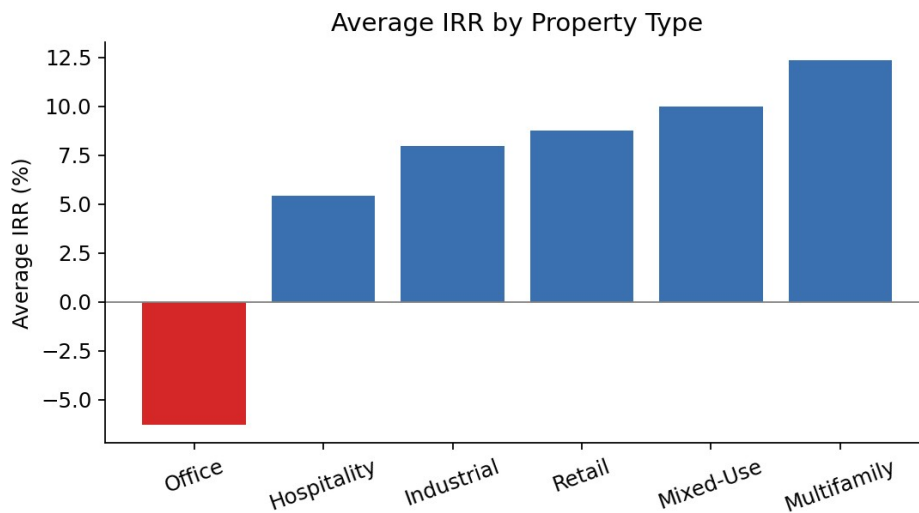


Figure 2. Office is the only property type with a negative average IRR, driven by cap rate expansion since acquisition — not an isolated model artifact. Public office REITs (BXP, VNO) have faced the same well-documented valuation pressure since 2022 (see [reports/reit_reference_benchmarks.md](#)), while industrial and multifamily have remained comparatively resilient — both in this portfolio and in the real public market.

5. Recommendation Mix & Value Creation

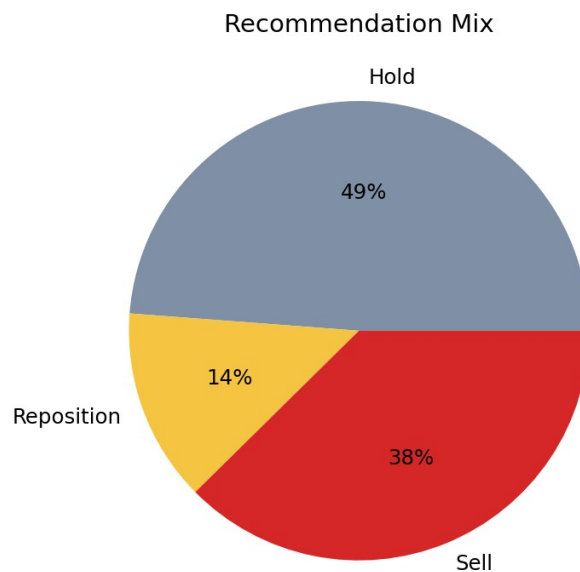


Figure 3. 48.8% Hold, 37.6% Sell, 13.6% Reposition. Read “Sell” carefully — it captures two structurally different cases that should not be treated as one signal:

- Harvest sale: IRR > 16% and equity multiple > 1.8x — a stabilized winner where the recommendation is to realize gains, not that anything is wrong.
- Distressed exit: IRR < 0% or interest coverage < 1.1x — a genuinely underperforming asset where continued holding is the risk.

The value-creation score (0–100) combines two ingredients in a stated, transparent formula — not a black box: 50% the value already captured (cap rate compression since acquisition) and 50% the model-forecasted forward NOI growth. Average portfolio score: 50/100, as designed — it is a percentile-based ranking, so the portfolio average is definitionally centered.

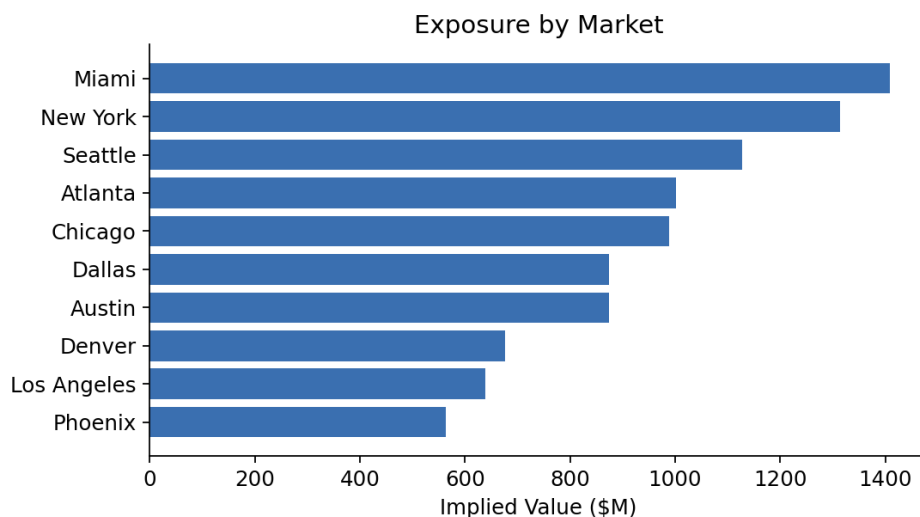


Figure 4. Exposure by market, current implied value.

6. Scenario Analysis — Moderate Downside

Scenario applied: -5% rent/NOI growth, -8% occupancy, +75bps exit cap rate, +15% capex.

Metric	Baseline → Shocked
Median portfolio IRR	8.3% → -8.8%
Impaired properties (negative or undefined IRR)	66 → 186 (of 250)
Implied portfolio value	\$9,469M → \$7,433M

A note on “undefined” IRR, because it matters and should not be glossed over: under stress, a meaningful number of properties become so underwater at exit — debt balance exceeding the shocked sale value — that no discount rate makes the cash flows break even. This is not a missing data point; it is a worse outcome than a very negative IRR, and this platform counts it as such rather than silently excluding it from the impaired count. An earlier build of this tool made exactly that mistake — caught and corrected before this memo was finalized — and is documented in the Model Card as a worked example of why every metric in an automated system needs a defined edge-case behavior, not just a happy-path calculation.

Under a moderate downside, the share of the book in genuine distress moves from roughly one in four properties to nearly three in four — the kind of shift a static, point-in-time review would not surface until it had already happened.

7. Limitations & Governance Recommendations

This system is built to structure the investment committee's discussion — not to replace it. Specific limitations:

- Synthetic ground truth: forward NOI growth predictions are validated against simulated, not observed, outcomes. Retraining against real NOI trajectories is required before production use.
- One-year forecast horizon only — no multi-year pro forma or full development timeline is modeled.
- Small sample size: 250 properties, ~187 in the forecast model's training fold; treat performance estimates as directional.
- Qualitative blind spots: sponsor/manager quality, submarket development pipeline, and tenant credit are invisible to a model trained on financial ratios alone.
- Scenario modeling simplification: hold period is held constant under shock (an “if these conditions applied through an exit today” view), not a re-solved multi-year forward pro forma; debt terms are held fixed.

Recommendation: use this platform to prioritize which assets receive deeper underwriting attention each quarter, particularly the office concentration and any property flagged impaired under the moderate downside scenario — not as a standalone buy/hold/sell signal.

8. Conclusion & Deliverables

This project demonstrates an end-to-end real estate equity analytics workflow: a genuine cash-flow-based return model (not assigned figures), unsupervised segmentation, an honestly-evaluated forecasting model, live scenario analysis, and governance-aware documentation — including a real bug found and corrected during development, disclosed rather than hidden, which is itself part of what a rigorous analytics practice looks like.

Full deliverable set:

- Live Streamlit scenario application
- Power BI asset-management dashboard
- GitHub repository with documented pipelines
- Data Dictionary
- Model Card
- This Investment Committee Memorandum
- AI extraction & verification report
- Demonstration video
- LinkedIn case study